Algebraic Geometry Problem Set #2

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Problem (Exercise 1). *Let $X \subseteq \mathbb{A}^n$ be an affine algebraic set endowed with an action of a finite group G. Let $A = \mathcal{O}(X)$ be the corresponding algebra, so that G acts on A by k-algebra automorphisms (i.e., $(g \star f)(x) = f(g^{-1}x)$ for $f \in A$). Let*

$$B = A^G = \{f \in A : g \star f = f \text{ for all } g \in G\} \subseteq A$$

be the subring of invariants. Last semester, we deduced from the Artin-Tate lemma that B is a finitely generated k-algebra. This allows us to consider $Y = MSpec(B)$, the affine algebraic set with $\mathcal{O}(Y) = B$. Let $\pi : X \to Y$ be the morphism induced by the inclusion $B \hookrightarrow A$. Show that

(a) $Y = X/G$ is the orbit space (that is, $\pi(x) = \pi(x')$ iff $x' = gx$ for some $g \in G$);

(b) for every morphism $f : X \to Z$ to an affine algebraic set Z such that $f \circ g = f$ for every $g \in G$ there exists a unique $\overline{f} : Y \to Z$ such that $f = \overline{f} \circ \pi$.

Solution.

Part (a). We need to first understand what condition $\pi(x) = \pi(x')$ even means. Note that each point $x \in X$ corresponds to the maximal ideal $\mathfrak{m}_x \subseteq A$, and so the morphism π sends $\mathfrak{m}_x \mapsto \mathfrak{m}_x \cap A^G$. Thus, $\pi(x) = \pi(x')$ means that $f(x) = f(x')$ holds for every $f \in A^G$.

Now we prove two directions. First assume $x' = gx$. Take any $f \in A^G$. Then, $f(x') = f(gx) = (g^{-1} \star f)(x) = f(x)$ (the last equality follows from invariance).

The remaining direction is much less trivial. Assume $x, x' \in X$ are two points with disjoint orbits Gx and Gx'. The idea of the remaining part of the proof is to construct some carefully chosen invariant $F \in A^G$ such that $F(x) \neq F(x')$.

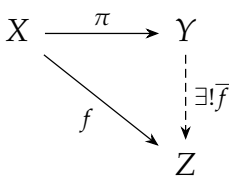
Since G is finite, both orbits are closed sets in the Zariski topology. Consequently, we can construct the regular function $s \in A$ such that

$$s(z) = \begin{cases} 1, & z \in Gx, \\ 0, & z \in Gx', \\ \text{whatever in the remaining points.} \end{cases}$$

Our candidate then is $F := \prod_{g \in G}(g \star s)$. Check all criteria:

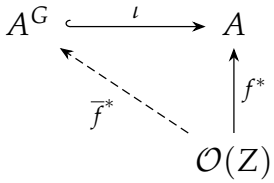
1. $\underline{F \in A^G}$: indeed, $g \star F = g \star \prod_{h \in G} h \star s = \prod_{h \in G} g \star h \star s = \prod_{h \in G}(gh) \star s$. Now note that the function $h \mapsto gh$ is a bijection, thus by reindexing the product we obtain the desired: $g \star F = F$.
2. $\underline{F(x) = 1}$: note that $F(x) = \prod_{g \in G} s(g^{-1}x) = \prod_{y \in Gx} s(y) = 1$.
3. $\underline{F(x') = 0}$: similarly $F(x') = \prod_{g \in G} s(g^{-1}x') = \prod_{y' \in Gx'} s(y') = 0$.

Part (b). We need to check the following universal property:



Here, the morphism $f : X \to Z$ satisfies $f \circ g = f \; \forall_{g \in G}$.

Recall that this morphism induces the k-algebra homomorphism between the coordinate rings $f^* : \mathcal{O}(Z) \to \mathcal{O}(X)$, thus defining the diagram:



Now let us consider the image of f^*. Take any $\phi \in \mathcal{O}(Z)$ and $x \in X, g \in G$. Then,

$$(g \star f^*(\phi))(x) = f^*(\phi)(g^{-1}x) = \phi(f(g^{-1}x)) = \phi(f(x)) = f^*(\phi)(x).$$

Thus, $g \star f^*(\phi) = f^*(\phi)$ for all $g \in G$ and thus $f^*(\mathcal{O}(Z)) \subseteq A^G$. Since the image factors through A^G, f^* uniquely factors as $\mathcal{O}(Z) \xrightarrow{\overline{f}^*} A^G \hookrightarrow A$. Finally note that we can pull back from statement $f^* = \iota \circ \overline{f}^*$ to $f = \overline{f} \circ \pi$, which was needed to prove.

Problem (Exercise 2). *Let $k = \mathbb{C}$. For which values of the parameter λ is the following hypersurface $X \subseteq \mathbb{P}^n$ non-singular?*

$$\lambda(X_0^{n+1} + \cdots + X_n^{n+1}) = (n+1)X_0 \dots X_n$$

Solution. We apply the Jacobian criterion for $\mathbb{P}^n$. Note that $X = V(f)$ where

$$f(X_0, \dots, X_n) = \lambda \sum_{j=0}^{n} X_j^{n+1} - (n+1) \prod_{j=0}^{n} X_j.$$

Find partial derivatives:

$$\frac{\partial f}{\partial X_i}(X_0, \dots, X_n) = \lambda(n+1)X_i^n - (n+1) \prod_{j \neq i} X_j.$$

Thus, all singular points $(X_0 : \dots : X_n)$ must satisfy:

(a) $\lambda \sum_{j=0}^{n} X_j^{n+1} = (n+1) \prod_{j=0}^{n} X_j$;

(b) $\lambda X_i^n = \prod_{j \neq i} X_j$ (for all $0 \leq i \leq n$).

Note that condition *(b)* always implies condition *(a)*. Indeed,

$$\lambda \sum_{j=0}^{n} X_j^{n+1} = \sum_{j=0}^{n} X_j \cdot (\lambda X_j^n) = \sum_{j=0}^{n} X_j \prod_{k \neq j} X_k = \sum_{j=0}^{n} \prod_{k=0}^{n} X_k = (n+1) \prod_{j=0}^{n} X_j.$$

Thus it suffices to consider only condition *(b)*. Notice that diving i^{th} equation by the first yields $X_i^n / X_0^n = X_0 / X_i$. Consequently, $(X_i / X_0)^{n+1} = 1$ or $X_i = \zeta_i X_0$ for $(n+1)^{\text{th}}$ root of unity ζ_i. Substituting this to any of equations, one gets:

$$\lambda X_0^n \zeta_i^n = \prod_{j \neq i} X_0 \zeta_j \Rightarrow \lambda = \prod_{j=0}^{n} \zeta_j.$$

Since product of roots of unity is again a root of unity, so is λ. Consequently, X is non-singular iff $\lambda^{n+1} \neq 1$.

Answer. λ must not be the $(n+1)$th root of unity nor 0.

Problem (Exercise 3). *Let X be a variety and let $X_1, \ldots, X_n \subseteq X$ be constructible subsets such that $X = \bigcup_{i=1}^n X_i$. Show that one of the X_i contains a non-empty open subset of X.*

Solution. By taking the closure of $X = \bigcup_{i=1}^n X_i$, we obtain $\overline{X} = X = \bigcup_{i=1}^n \overline{X}_i$. Since X is an irreducible topological space, it cannot be covered by a finite union of proper closed subsets. Therefore, at least one of such sets is dense: say, $\overline{X}_k = X$. Since X_k is constructible, we know that $X_k = \bigcup_{i=1}^m L_i$ where $L_i = U_i \cap C_i$ (U_i is open and C_i is closed). Again, by taking the closure, we obtain $X = \bigcup_{i=1}^m \overline{L}_i$. Applying the same argument once more, one of L_i is dense: say, $\overline{L}_r = X$. But now notice the following three facts:

1. Since $U_r \cap C_r \subseteq C_r$, we have $\overline{L}_r \subseteq C_r$.
2. Since $\overline{L}_r = X$, we have $X \subseteq C_r$, so $X = C_r$.
3. Consequently, $L_r = U_r \cap C_r = U_r \cap X = U_r$.

Thus, our sought non-empty open subset of X is precisely L_r (note that it is non-empty since L_r is both dense and open).

Problem (Exercise 4). *Let $X = V(f) \subseteq \mathbb{P}^2$ be a conic (so $f \in k[X, Y, Z]$ is an irreducible homogeneous polynomial of degree 2). Show $X \cong \mathbb{P}^1$.*

Solution. We prove this fact in two steps:

- **Step 1.** $X \cong V(X^2 + Y^2 + Z^2) =: C$ (*i.e., all conics are isomorphic, so we do not care which one to pick*);
- **Step 2.** $C \cong \mathbb{P}^1$ (*seems easiest to show this fact for $X^2 + Y^2 + Z^2 = 0$*).

Step 1. Letting $\mathbf{X} = (X, Y, Z)^\top$, write f as a quadratic form $f(\mathbf{X}) = \mathbf{X}^\top A \mathbf{X}$. Now notice the following: since A is a symmetric matrix, A can be diagonalized (excluding the case $\mathrm{char}(k) = 2$). Therefore, there exists the linear transformation $\mathbf{X} \mapsto P\mathbf{X}$ (for some $P \in k^{3\times 3}$) such that

$$f(P\mathbf{X}) = \mathbf{X}^\top D \mathbf{X}, \quad D = \begin{bmatrix} \lambda_X & 0 & 0 \\ 0 & \lambda_Y & 0 \\ 0 & 0 & \lambda_Z \end{bmatrix}.$$

Moreover, since f is irreducible, $\lambda_X, \lambda_Y, \lambda_Z \neq 0$. Therefore, by applying $(X : Y : Z) \mapsto (X/\sqrt{\lambda_X} : Y/\sqrt{\lambda_Y} : Z/\sqrt{\lambda_Z})$, we deduce $X \cong V(X^2 + Y^2 + Z^2)$.

Remark. Can it so happen that either of λ_X, λ_Y, or λ_Z is 0? Notice that $f(X, Y) = \lambda_X X^2 + \lambda_Y Y^2$ is clearly reducible: $f(X, Y) = (\sqrt{\lambda_X} X + i\sqrt{\lambda_Y} Y)(\sqrt{\lambda_X} X - i\sqrt{\lambda_Y} Y)$. Thus, we need to have at least three variables for f to be irreducible.

Step 2. Define the map $\phi : \mathbb{P}^1 \to C$ as

$$(U : V) \mapsto (U^2 - V^2 : 2UV : i(V^2 + U^2)),$$

where $i^2 = -1$. One easily checks that this map is an isomorphism: first, clearly

$\phi(\mathbb{P}^1) \subseteq C$ (substitute into equation of C and simplify). Next, one checks that the inverse map is given by $\psi : C \to \mathbb{P}^1, (X : Y : Z) \mapsto (X + iY : Z)$ and is regular.

Problem (Exercise 5). *Assume $\mathrm{char}(k) \neq 2$. Let $X = V(f) \subseteq \mathbb{P}^3$ be an irreducible quadric surface (so $f \in k[T_0, \ldots, T_3]$ is an irreducible homogeneous polynomial of degree two). Show that X is rational.*

Solution. As the hint suggests, we first show the following.

Proposition. X is isomorphic to either a surface $V(T_0T_1 - T_2T_3)$ or $V(T_0T_1 - T_2^2)$.

Proof of proposition. Using the same argument as in *Exercise 4*, any quadric is isomorphic to either $T_0^2 + T_1^2 + T_2^2 + T_3^2 = 0$ or $T_0^2 + T_1^2 + T_2^2 = 0$ (the one- and two-degree cases will trivially result in reducible polynomial f).

Using the following linear change of variables for the four-degree case:

$$T_0 := \frac{U+V}{2}, \quad T_1 := \frac{U-V}{2i}, \quad T_2 := \frac{W+Z}{2}, \quad T_3 := \frac{W-Z}{2i},$$

one obtains exactly equation $UV - ZW = 0$.

For the three-degree case the substitution is even simpler:

$$T_0 := \frac{U+V}{2}, \quad T_1 = \frac{U-V}{2i}, \quad T_2 := iW,$$

resulting in the equation $UV - W^2 = 0$.

Proof of Exercise. First consider the case $X \cong V(T_0T_1 - T_2^2)$. We shall build the birational map $\mathbb{P}^2 \dashrightarrow X$. We propose the following map:

$$(U : V : W) \mapsto (U^2 : V^2 : UV : VW).$$

To prove it is birational, we need to show that it admits the rational inverse $X \dashrightarrow \mathbb{P}^2$. But this map is very simple: $(T_0 : T_1 : T_2 : T_3) \mapsto (T_2 : T_1 : T_3)$.

Now assume $X \cong V(T_0T_1 - T_2T_3)$. Here, the map $\mathbb{P}^2 \dashrightarrow X$ will be defined as:

$$(U : V : W) \mapsto (UW : VW : UV : W^2).$$

The rational inverse $X \dashrightarrow \mathbb{P}^2$ is then given by

$$(T_0 : T_1 : T_2 : T_3) \mapsto (T_0 : T_1 : T_3).$$